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Deep supervised anomaly detection for generalized face forgery detection

Fan Zhang [a,b], Ting Yang [a], Lin Cao [a,b,*], Kangning Du [b], Yanan Guo [a], Peiran Song [b], Chen Shao [a]

[a] Key Laboratory of Information and Communication Systems, Ministry of Information Industry, Beijing Information Science and Technology University (BISTU), Beijing 100101, PR China

[b] Key Laboratory of the Ministry of Education for Optoelectronic Measurement Technology and Instrument, BISTU, Beijing 100101, PR China

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ABSTRACT

Nowadays, face forgery poses a significant threat to societal security, making the development of effective countermeasures imperative. Though most existing methods adopt neural networks to automatically extract discriminative features for forgery detection and have achieved promising results, significant challenges remain. Namely, when detecting forgery faces generated by unseen forgery methods, the detection performance degrades significantly, indicating poor generalization capability. To address such limitation, a novel deep supervised anomaly detection for generalized face forgery detection (DAGFD) is proposed in this paper. Specifically, the artifact map detector optimized by triplet focal loss and metric-softmax loss is first used to locate the forgery regions and obtain artifact maps. Next, forgery detection is reformulated from the supervised anomaly detection perspective, and the artifact map score is calculated to detect forgery videos. Furthermore, mean square error (MSE) loss is used to minimize the artifact map score of real samples while increase the one of forgery samples to generalize well to unseen forgery methods. Also, circle loss is used for auxiliary classifier to learn more discriminative artifact features. Finally, the experimental results demonstrate that the proposed method's detection accuracy is better than other state-of-the-art methods.

1. Introduction

In recent years, the short video platforms have developed rapidly, and numerous videos of celebrities and ordinary people have emerged on the Internet. Meanwhile, the face videos can easily be manipulated by face forgery techniques to generate fake videos, which are too realistic to be distinguished by human eyes, as shown in Fig. 1. Supposing the technology is used in a specific scene with malicious purposes, such as false news, celebrity scandal, and pornographic videos, it will bring severe trust crises and potential safety hazards to society. To mitigate such risks, various face forgery detection approaches have been proposed.

Early traditional approaches utilized hand-crafted features [1], such as frequency domain features and image noise characteristics, to detect face forgery videos. However, these methods were less effective in discriminating face forgery videos generated by deep learning methods. Afterwards, with the tremendous success of deep learning, some works adopted convolutional neural network (CNN) to automatically extract discriminative features for forgery detection [2]. For example, Afchar et al. [3] proposed MesoNet with fewer layers for forgery detection, which focused on the details of the fake image. Rössler et al. [4] found that the features extracted by the deeper network were more discriminative, so 2048-dimensional features were extracted from the face region via XceptionNet. Compared with the traditional methods, CNN-based methods have achieved significantly improved detection accuracy.

However, when facing challenging forgery videos, the performances of the above-mentioned methods are not satisfactory. For this purpose, Akash et al. [5] demonstrated that the use of metric loss can effectively improve the detection accuracy of CNN models. Li et al. [6] proposed a novel frequency-aware discriminative feature learning framework with metric learning to enhance inter-class separation in the embedding space. In addition, there are methods focusing on localizing the forgery region to explore the widespread artifacts in forgery samples. For example, Zhao et al. [7] formulated forgery detection as a fine-grained classification problem and proposed a multi-attentional forgery detection network, where multiple spatial attention heads and textural feature enhancement blocks were utilized to zoom in the subtle artifacts. Nguyen et al. [8] adopted multi-task learning and used encoder and Y decoder to segment and reconstruct image tampering

* Corresponding author at: Key Laboratory of Information and Communication Systems, Ministry of Information Industry, Beijing Information Science and Technology University (BISTU), Beijing 100101, PR China.

E-mail address: charlin26@163.com (L. Cao).

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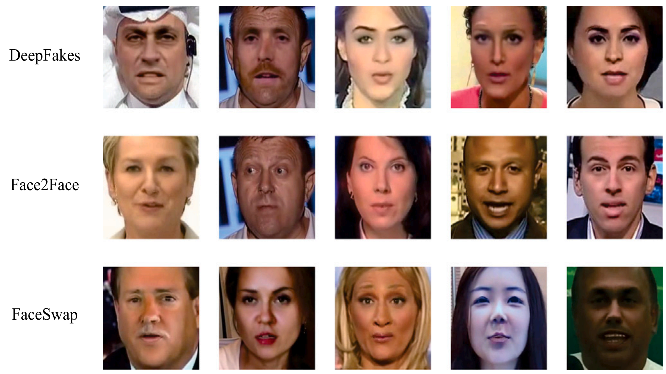

Fig. 1. Forgery images.

area, achieving superior results. Rather than simply using a multi-task learning strategy, Nicolo et al. [9] integrated several CNNs, and the attention mechanism combined with metric learning was used to assist the model training. Although the detection methods of using metric learning and localizing the forgery region have achieved good results, significant performance degradation is observed when applying the trained models to unseen forgery types [5]. Namely, the generalization capability is poor. The reason for the poor generalization is that the detection model cannot extract common features from the forgery videos generated by different generation methods to distinguish fake videos from the real ones. That is, because there are diverse generation methods with different semantic information, it is difficult to obtain an obvious decision boundary in the feature space for real and forgery videos.

In summary, when existing detection methods detect face forgery videos generated by unseen forgery methods, the detection performance decreases seriously. To address such limitation, this paper proposes deep supervised anomaly detection for generalized face forgery detection (DAGFD), as shown in Fig. 2. Specifically, the artifact map detector (AMD) is first applied to locate the forgery region and obtain artifact maps. Then, face forgery detection is reformulated from the supervised anomaly detection perspective, where the artifact map of the fake video is assumed to be an all-one mask, while the one of the real video is an all zero mask. Next, artifact map scores are computed using artifact maps optimized by the MSE loss to determine whether the video is real or fake. The optimization aims to minimize the scores of real samples on closed-set and increase the ones of fake samples on open-set. Furthermore, metric-softmax loss combining with triplet focal loss is used to enhance the discriminability of the artifact map. And circle loss used for auxiliary classifier benefits feature learning with high flexibility in optimization. Finally, the experimental results demonstrate that the proposed method's detection accuracy is better than state-of-the-art comparison methods.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 describes the proposed method, including the overall network structure, supervised anomaly detection, and loss functions. Section 4 shows the experimental results, concluding remarks and future works are presented in the final section.

2. Related work

In this section, we briefly review face forgery detection methods.

2.1. Methods based on traditional hand-crafted features

Bestagini et al. [1] demonstrated that forgery videos anomalously disrupted the encoding footprints originating from capture devices, a disruption distinct from the modifications caused by normal re-encoding. This finding motivated their proposed face forgery detection method based on analyzing these footprints. Matern et al. [10] compared the head pose errors between real and fake faces, and then support vector machine (SVM) was used for classification. Agarwal et al. [11] extracted features from five facial sensory organs for forgery detection. However, the model can only use one person's video for training, so it can only detect the specific person's video with low efficiency. In brief, traditional methods extract the physical features to detect forgery videos. Consequently, these methods are ineffective in distinguishing forgery videos generated by deep learning methods.

2.2. Methods based on deep learning

Methods based on deep learning have achieved improved detection accuracy compared to traditional detectors [12]. Bayar et al. [13] put forward a network architecture with the new convolution layer form, which can directly identify the features caused by forgery methods. The method proposed by Rahmouni et al. [14] used a self-defined pooling layer, where four kinds of statistics (mean, variance, maximum and minimum) were calculated to predict whether the face image was fake or not. Rössler et al. [4] extracted features from the face region via XceptionNet for forgery detection.

The performances of early deep learning based methods to detect challenging forgery videos are not ideal, so some researchers try to improve the detection accuracy by locating the forgery areas. Nguyen et al. [8] used multi-task learning to locate the forgery areas of fake videos. The model including the encoder and Y decoder was designed, where one branch of the decoder performed segmentation of the forgery area, and the other branch reconstructed the input. Bonettini et al. [9] simultaneously used the EfficientNetB4, the attention mechanism and improved Siamese network, achieving good results on two public datasets. However, due to the model complexity and high training cost, it had not been widely used. Li et al. [15] noted that the forgery process of most forgery techniques had the step of fusing the target face with the original background, thus remaining artifacts at the fusion boundary. Therefore, a mixed boundary graph was generated to assist model training, thus forcing the model to identify the fusion region of fake videos, which achieved superior generalization. Chen et al. [16] formulated DeepFake detection as multi-task learning problem to simultaneously predict the intrinsic and extrinsic artifacts.

Moving beyond single-frame analysis, some researchers utilize inter-frame features of video for detection [17]. D. Güera et al. [18] proposed a CNN-RNN architecture for face forgery detection. The CNN adopted InceptionV3 [19] pre-trained on ImageNet [20], and the temporally-aware RNN focused on capturing temporal inconsistencies between frames introduced by the face swapping process. Pu et al. [21] proposed a dual-level collaborative framework to detect frame-level and video-level forgeries with a joint loss function to optimize both the AUC score and error rate simultaneously. Nie et al. [22] proposed the transformer-based framework for diffusion learning of temporal inconsistency pattern (DIP), which exploited directional inconsistencies for deepfake video detection. The methods of utilizing inter-frame features require large datasets and complex models, which are not suitable for practical applications.

Different from the above-mentioned methods, several studies adopted multiple loss fusion to improve generalization ablity. Sun et al. proposed dual contrastive learning for general face forgery detection [23], where both cross-entropy loss based supervised learning branch and two InfoNCE losses based contrastive learning branches were utilized. The work in [24] proposed multimodal contrastive learning for Deepfake detection, where seven diverse loss functions were utilized to explore intra-modal and cross-modal forgery clues. Zhang et al. [25] proposed bi-source reconstruction-based classification network for forgery detection, where reconstruction loss for amplifying forgery clues, metric loss for measuring image distances, and binary-cross entropy loss for classification were applied. Huang et al. [26] designed explicit identity contrast (EIC) loss and implicit identity

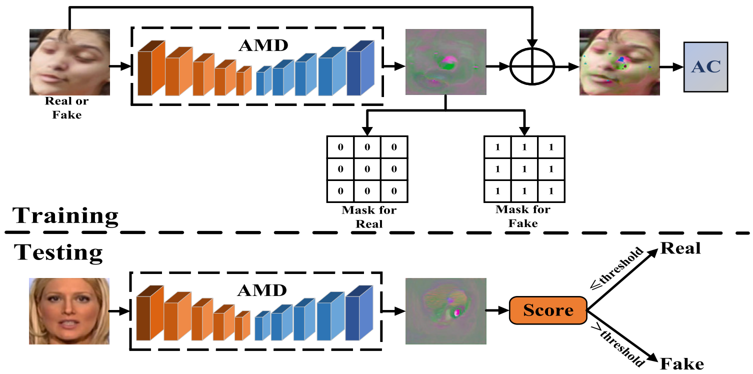

Fig. 2. Overview of the DAGFD training and testing procedures. In the training phase, artifact map detector (AMD) is first applied to input images to obtain artifact maps. Then, the artifact map of real image is expected to approach the all-zero mask, while the one of forgery image approaches the all-one mask. Finally, the superimposition of artifact map and input is input into an auxiliary classifier. During the testing phase, the input is processed through AMD to obtain artifact map, and artifact map score is computed to judge whether the image is real or fake.

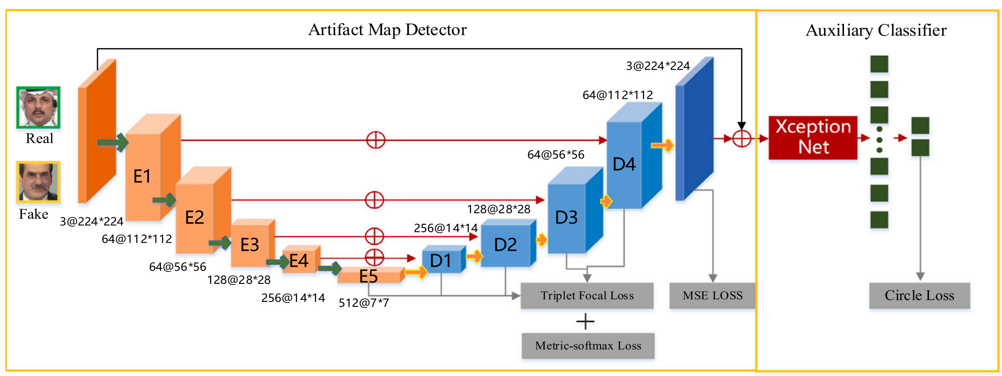

Fig. 3. The proposed network architecture.

exploration (IIE) loss, which supervised a CNN backbone to embed face images into the implicit identity space for face swapping detection. However, the above multiple loss fusion methods typically rely on classfiers for forgery detection, which is different from artifact map score calculation in our proposed method.

2.3. Methods based on anomaly detection

Andrea et al. [27] proposed Deepfake detection method exploiting surface anomalies, where global surface descriptor (GSD) was used as feature that accounted for the camera acquisition process which marked permanently the image. The work in [28] considered the feature anomalies of forged faces caused by the prevalent blending operations in face forgery algorithms, proposing a weakly supervised second order local anomaly (SOLA) learning module to mine anomalies in local regions using deep feature maps. Oliver et al. [29] proposed a pipeline capable of detecting GAN specific frequencies (GSF) by employing discrete cosine transform (DCT). Their analysis revealed that β statistics inferred by the AC coefficients distribution were the key to recognize GAN-engine generated data. Wang et al. [30] proposed face depth map transformer (FDMT) to estimate the face depth map patch by patch from a RGB face image, enabling the capture of local depth anomaly created due to manipulation. Zhang et al. [31] proposed an effective sequence-based forgery detection framework based on an existing video classification method. To make the learned features more generalizable, an auxiliary anomaly detection block was proposed. During inference, the classifiers of the backbone and the anomaly detection branch both had a forgery prediction, and the average of them were used for forgery detection. However, these anomaly detection based methods use classifiers to determine whether an image has been forged, hence the generalization performance is still limited.

In summary, when detecting face forgery videos generated by unseen forgery methods, the performances of existing detection methods degrades significantly. In response to the above problem, the paper proposes DAGFD to detect face forgery videos from the perspective of supervised anomaly detection. Specifically, the real samples are assumed to belong to a closed-set while the fake samples are outliers of the closed-set and belong to an open-set, and then the artifact map scores of real samples are minimized while the ones of forgery samples are increased to generalize to unseen forgery methods. It is worth noting that the classifier in DAGFD is utilized as an auxiliary module, and the artifact map score is calculated to judge whether the video has been forged, which is different from above-mentioned methods that use classifiers for forgery detection.

3. Method

3.1. The overall framework

Motivated by the residual learning, the differences between real and fake faces can be regarded as the residual of the real and fake faces. Based on this, the difference is defined as the artifact map assuming the artifacts only exist in fake faces. Therefore, artifact map detector (AMD) combined with auxiliary classifier (AC) is proposed in a residual learning manner. The proposed DAGFD is depicted in Fig. 3, where artifact map detector (AMD) is utilized to obtain the artifact map and locate the forgery regions of fake images, and auxiliary classifier (AC) with circle loss (CL) assists AMD module to learn more discriminative artifact features. Specifically, the image I is first input into AMD to obtain the artifact map C. Next, C is added to the input image I to form the overlapped image S, and feed S into the AC to make the learned artifact map more discriminative. Furthermore, artifact map score optimized by MSE loss is computed to judge whether the input I is real or fake.

AMD: The AMD aims to locate the forgery regions and amplify the artifact features of the forgery videos, where the U-Net architecture, building skip connections from encoder to decoder in multiple scales to obtain forgery cues, is adopted. Specifically, the ResNet18 pre-trained on ImageNet is chosen as encoder E which contains five encoder residual blocks $E_1 - E_5$. Followed by E, the decoder D, consisting of four decoder residual blocks $D_1 - D_4$, decodes the information to obtain the artifact map C. In each decoder residual block, the feature maps from the previous layer are upsampled by a nearest-neighbor interpolation, followed by 2×2 convolution. Similar to the residual learning method, the feature maps of the corresponding encoder are fused with the output of the decoder. Typically, the input of encoder $E1$ is a real or fake face image I, and the input of $E2 - E5$ is the output of the upper encoder. Meanwhile, the input of $D1$ decoder is the concatenation of the output of $E5$ and $E4$ encoder, and the input of $D2 - D4$ is the concatenation of the output of the upper layer decoder and the corresponding encoder. Finally, the output of D4 is the artifact map with 3 channels. Given that the input image consists of 3 channels (RGB), maintaining the artifact map in the same format ensures dimensional consistency for subsequent addition operation.

Auxiliary classifier (AC): Feng et al. [32] demonstrated that combining metric learning based loss with classification supervision facilitates learning more discriminative features, which inspires us to design the auxiliary classifier. In proposed framework, auxiliary classifier is used to assist the AMD module, which is optimized by circle loss. Specifically, XceptionNet [33] is used as the feature extraction network for the auxiliary classifier, where the channel and spatial correlations are completely separated. Namely, XceptionNet performs spatial convolution to map the spatial correlation of each channel individually, followed by 1×1 convolution to map the channel correlation. As shown in Fig. 3, add artifact map C and input image I to form the overlapped image S in a residual learning manner, and S is input into the XceptionNet for extracting 2048-dimensional features x_{2048}. Finally, x_{2048} is mapped to a 2-dimensional feature x_2 via a fully connected layer.

3.2. Supervised anomaly detection

Anomaly detection is the task of identifying the unusual samples in a standard set. The objective of anomaly detection [32] is to train the neural network $\phi(x; w)$ to learn a transformation that minimizes the volume of a data-enclosing hypersphere in output space z centered on a predetermined point c. Namely, given N unlabeled training samples ($x_1, \ldots, x_{N1} \subseteq X$), the objective function is:

$$\min_{w} \frac{1}{N} \sum_{i=1}^{N} \left\| \phi\left(x_i; w\right) - c \right\|^2 + \frac{\lambda}{2} \sum_{i=1}^{N} \left\| w^{\ell} \right\|_F^2, \lambda > 0 \tag{1}$$

where ϕ is the transformation learned by the neural network, c is the center point of the closed hypersphere, N is the number of training samples, w is the weight, l is the number of hidden layers, and λ is the hyper-parameter.

The anomaly score for the test point x is given by the distance from $\phi(x; w)$ to the center of the hypersphere [34]:

$$S(x) = \|\phi(x; w) - c\|. \tag{2}$$

Within the field of forgery detection, real samples are assumed to share common characteristics, while fake samples exhibit significant diversity due to the variety of forgery generation methods. In light of this, decision boundary learned by traditional supervised learning may struggle to generalize to unseen forgery types. Therefore, supervised anomaly detection is introduced into the field of face forgery detection, aiming to improve the detection performance for unseen forgery types. Traditional anomaly detection models are typically trained in an unsupervised manner using only normal samples, which may result in ambiguous decision boundary and insufficient discriminability. However, forgery (anomaly) samples are often available in real-world applications, and the valuable knowledge of seen forgery samples should be effectively utilized. While incorporating seen forgery samples into training can improve detection, it also introduces the risk of model bias towards seen anomalies, potentially reducing generalization to unseen forgery types. To mitigate above issues, the proposed method adopts supervised anomaly detection to exploit the diversity, which constrains the artifact map of forgery sample to all-one mask regardless of whether the forgery sample is seen or not. In this way, the bias problem caused by seen forgery samples can be alleviated, and better generalization performance is obtained.

Typically, given N_l real samples($x_1^r, \ldots, x_{N_l}^r \subseteq X$) and N_S fake samples ($x_1^f, \ldots, x_{N_s}^f \subseteq X$), let c be the center of the real samples in the output space z, the objective function is formulated as:

$$\min_{w} \frac{1}{N_l} \sum_{i=1}^{N_l} \left\| \phi\left(x_i; w\right) - c \right\|^2 \tag{3}$$

$$\max_{w} \frac{1}{N_s} \sum_{i=1}^{N_s} \left\| \phi\left(x_i; w\right) - c \right\|^2 \tag{4}$$

The proposed method assumes the mask of the real sample is all-zero matrix, while the one of the fake sample is all-one matrix. Then, the mask and the artifact map of the real or fake sample are input into MSE loss to achieve the optimization goal of Eqs. (3) and (4).

3.3. Loss functions

Four different loss functions are used in proposed method. Specifically, metric-softmax loss combining with triplet focal loss enhances the discriminability of the artifact map. Then, circle loss used for auxiliary classifier benefits feature learning with high flexibility in optimization. Finally, MSE loss is used to minimize the artifact map score of real samples while increase the one of forgery samples to generalize well to unseen forgery methods.

3.3.1. Triplet focal loss

The triple loss, enhancing inter-class separability and intra-class compactness, is shown in Eq. (5):

$$L_t = \frac{1}{T} \sum_{i=1}^{T} max\left(d\left(a_i, p_i\right) - d\left(a_i, n_i\right) + m, 0\right) \tag{5}$$

where $\{a_i, p_i, n_i\}$ denotes the anchor (real), positive (real), negative (fake) samples within the ith triplet, respectively, T is the number of triplets, $d(i, j)$ represents the Euclidean distance of $L2$-normalized feature vectors output by the AMD module, and m is the pre-defined margin.

However, the triple loss mentioned above treats all examples equally, which causes these methods to treat hard examples like easy examples. Therefore, triplet focal loss, paying more attention to hard negative samples, is applied to train the AMD. Specifically, triplet focal loss maps Euclidean distance to the exponential kernel, and hence the weight of hard negative samples can be automatically increased. That means increasing the penalty factor of hard fake samples, thereby improving model performance. In detail, the triplet focal loss, used for the artifact map detector's $E5$, $D1$, $D2$, $D3$, and $D4$ layers, is shown in Eq. (6):

$$L_{tf} = \frac{1}{T} \sum_{i=1}^{T} max\left(0, e^{\left(\frac{d\left(a_i, p_i\right)}{\sigma}\right)} - e^{\left(\frac{d\left(a_i, n_i\right)}{\sigma}\right)} + m\right) \tag{6}$$

where σ is the hyperparameter that controls the magnitude of the exponential kernel.

The goal of triplet focal loss is to select hard negative sample within a mini-batch. In each training step, a set of samples of each class is chosen. Then, the distances between all positive pairs are computed, i.e. $d(a,p)$, and the distance between anchor and all negative samples are computed, i.e. $d(a,n)$. Eventually, the negative sample is chosen based on the criteria $d(a,p) - d(a,n) < m$ to obtain the final tuples. In this way, all training samples are related to the learning process, and hence better performance is achieved.

3.3.2. Metric-softmax loss

The metric-softmax loss proposed by Pérez-Cabo et al. [35] improves softmax loss. Specifically, metric-softmax loss is used to accumulate the probability distribution within each triplet and make it highly separable in Euclidean space. The proposed method expects the features of the fake face images to be more representative in order to distinguish from real face images. Namely, the metric-softmax loss, as shown in Eq. (7), is adopted to optimize the AMD to further separate the fake samples from the real samples, making the artifact maps more discriminative.

$$L_{\mathrm{msl}} = -\sum_{i=1}^{b} \log \frac{e^{d_{a_i,p_i}}}{e^{d_{a_i,p_i}} + e^{d_{a_i,n_i}}} \tag{7}$$

where b is the number of triplets in each batch, $\{a_i, p_i, n_i\}$ are the anchor (real), the positive (real) and the negative (fake) samples within the ith triplet, respectively.

In our work, we do not focus on binary classification task, but rather obtaining discriminative artifact map to distinguish real samples from forgery samples. Therefore, metric-softmax loss combining with triplet focal loss is used to enhance the discriminability of the artifact map, where triplet focal loss acts as the regularizer. In this way, the learning process towards binary classification is alleviated, and the generalization ability can be improved. The validity will be experimentally verified in Section 5.5.2.

3.3.3. MSE loss

The proposed method assumes that the real samples belong to the closed set, while the fake samples are outliers and belong to the open set. Consequently, the artifact map of real sample is assumed to be a zero mask, while the one of forgery sample is a non-zero mask. Namely, zero mask and non-zero mask are set, and the zero mask is considered as the center of the real samples in the feature space, then the optimization goal of Eqs. (3) and (4) is achieved through the MSE loss and two masks. Compared to methods relying on scalar scores, pixel-wise supervision helps to learn more about the differences between real and forgery samples. Based on this, the network predicts 224 × 224 × 3 artifact maps instead of traditional scalar scores, and the proposed method simply sets all pixel values of zero mask to 0 and the ones of non-zero mask to 1. In this way, the artifact map of forgery sample is constrained to all-one mask regardless of whether the forgery sample is seen or not, hence improving generalization performance. Typically, the MSE loss is shown in Eq. (8):

$$L_{MSE} = \frac{1}{H \times W \times D} \sum_{i \in H, j \in W, k \in D} \left(B_{artifact(i,j,k)} - B_{target(i,j,k)}\right)^2 \tag{8}$$

where H, W, D denotes the height, width and depth of the artifact map, respectively, $B_{artifact}$ is the artifact map, and B_{target} is the target mask.

3.3.4. Circle loss

Both classification loss (e.g., cross-entropy loss) and metric loss (e.g., triple loss) aim to increase intra-class similarity and reduce inter-class similarity, among which classification loss focuses more on optimizing the similarity between samples and weight vectors during backpropagation, and metric loss focuses on optimizing the similarity between samples. Namely, both types of losses have drawbacks: lack of optimization flexibility and ambiguous convergence state. To this end, Sun et al. [36] proposed the circle loss. Assuming that S_p is the intra-class similarity score and S_n is the inter-class similarity score, the commonly used losses expect the model to be optimized in the direction of decreasing (S_n-S_p). That is, when a specific sample A is at the optimization boundary, the model will continue to increase S_p and decrease S_n at A due to the same penalty of S_p and S_n, and then the value of S_n is close to 0. It is obviously unreasonable to optimize based on the same penalty, and such optimization method violates the feature space separability. For this purpose, the circle loss gives different penalty weights a_n and a_p to different similarity scores, and the new decision boundary is $a_n S_n - a_p S_p = m$.

The circle loss is used to optimize the AC module in proposed method, as shown in Eq. (9). Specifically, the farther the similarity score is from the optimal value, the larger the weighting factor is. In contrast, if the similarity score is close to the optimal value, it should be moderately optimized. Therefore, in order to make the model classify the hard samples correctly, the use of circle loss encourages the model to assign a larger penalty factor to the hard samples and the smaller one to the easy samples, allowing the model to optimize the sample similarity adaptively.

$$L_{cl} = \log\left[1 + \sum_{j=1}^{L} \exp\left(\gamma \alpha_n^j \left(S_n^j - \Delta_n\right)\right) \sum_{i=1}^{K} \exp\left(-\gamma \alpha_p^i \left(S_p^i - \Delta_p\right)\right)\right] \tag{9}$$

where,

$$\begin{cases} \alpha_p^i = \left[O_p - S_p^i\right]_+, \\ \alpha_n^j = \left[S_n^j - O_n\right]_+, \end{cases} \tag{10}$$

$O_p = 1+m$, $O_n = -m$, $\Delta_p = 1-m$, $\Delta_n = m$, $[\bullet]_+$ is the operator guaranteed to be non-negative, γ is the scale factor, m is the relaxation margin, S_p is intra-class similarity score, K is the number of intra-class similarity scores, S_n is the inter-class similarity score, L is the number of inter-class similarity scores, a_n and a_p are the weight factors for S_n and S_p, respectively.

The decision boundary of circle loss is shown in Eq. (11), and the final optimization objective is $S_p^i > 1 - m$, $S_n^j < m$.

$$\left(S_n - 0\right)^2 + \left(S_p - 1\right)^2 = 2m^2 \tag{11}$$

The purpose of circle loss is to optimize S_p to 1 and S_n to 0, and the final decision boundary is circular. Hence, circle loss used for auxiliary classifier benefits feature learning with high flexibility in optimization, which is better than the linear decision boundary of classification loss, e.g. Softmax.

In summary, the total loss of DAGFD is obtained by weighted summation of MSE loss L_{MSE}, triplet focal loss L_{tf}, metric-softmax loss L_{msl}, and circle loss L_{cl}, as shown in Eq. (12),

$$L_{all} = \alpha_1 L_{MSE} + \alpha_2 \sum_{k \in \{E5-D4\}} L_{tf}^k + \alpha_3 L_{msl} + \alpha_4 L_{cl} \tag{12}$$

where k indexes the layer where the proposed method applies the triplet focal loss, α_1, α_2, α_3 and α_4 are weights to balance different losses, and $E5 - D4$ are the outputs of the fifth encoder and the first four decoders of the AMD.

The process of DAGFD algorithm is shown in Algorithm 1.

Algorithm 1: DAGFD

Input: The image I
Output: The label: 0 (forgery) or 1 (real)
For all training images do
1. Perform image augmentation and normalization on image I to obtain V;
2. Input V into the AMD to extract the features $F_{E_5} - F_{D_4}$ and obtain the artifact map C;
3. Add artifact map C and input image I to obtain the overlapped image S, and input S into XceptionNet to extract 2048-dimensional feature vector x_{2048};
4. Pass x_{2048} through the fully connected layer of the AC module to obtain the 2-dimensional feature vector x_2;
5. Calculate the total loss L_{all} (triple focal loss L_{tf} and metric-softmax loss L_{msl} for $F_{E_5} - F_{D_4}$, MSE loss L_{MSE} for artifact map C, circle loss L_{cl} for x_2), and back propagate to update the parameters of AMD and AC.

end for

Table 1
Type and quantity distribution of dataset I and II.

Type	Dataset I		Dataset II	
	Train	Test	Train	Test
Youtubu	190 000	14 000	27 000	6000
Deepfake	190 000	14 000	27 000	6000
Faceswap	190 000	14 000	27 000	6000
Face2Face	190 000	14 000	27 000	6000

4. Experimental setup

In this section, we first present the datasets and then introduce the overall experimental setup, followed by a description of the testing strategy.

4.1. Datasets

Dataset I is used to keep consistent with the dataset of comparative methods, and ablation experiments are conducted on dataset II, consisting of randomly selected images from FaceForensics ++ (FF++) [4] dataset. Typically, three forgery types are selected from the FF++ dataset for our experiments, namely DeepFake, FaceSwap, and Face2Face. All videos from this dataset with c23 compression are utilized to benchmark diverse detection methods. Furthermore, 1000 forgery videos are divided into the training set, test set, and validation set according to the ratio of 720:140:140. For dataset I, 270, 100, and 100 images are taken from each video in the training set, test set and validation set, respectively. Namely, the training set has a total of 190,000 images, and both the test set and validation set have 14 000 images.

To save the time cost of ablation experiments, 200 forgery videos are divided into the training set and test set according to the ratio of 1:1. For dataset II, 270 and 60 images are taken from each video in the training set and test set, respectively. Namely, the training set has a total of 27,000 images, and the test set has 6000 images. Table 1 summarizes the type and quantity distribution of datasets I and II.

4.2. Experimental settings

During training, the proposed method extracts $224 \times 224 \times 3$ face regions from each video frame. The batch size (BS) is set to 64, and the training epoch is 10. Adaptive Moment Estimation (ADAM) is used for optimization, and the initial learning rate is set to 0.0002. The parameter m for triplet focal loss and circle loss is set to 0.5 and 0.2, respectively. σ is 0.3, α_1, α_2, α_3 and α_4 is 5, 1, 1, and 5, respectively. The Pytorch is used as the basic framework, and the GPU is NVIDIA RTX 2080Ti.

Performance Evaluation: Accuracy (ACC) is used to measure the performance, defined by Eq. (13) :

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

where FP, FN, TN and TP denotes the number of false positive, false negative, true negative, and true positive samples, respectively.

Data Augmentation: The methods in the Albumentations library are utilized to augment and normalize the images, including gaussian noise and rotation, etc.

4.3. Test strategy

In the testing phase, the input image is processed through AMD to obtain artifact map. Whether the face image is real or fake is judged by calculating the artifact map score instead of the classifier output. Specifically, the artifact map score is obtained by calculating the pixel-wise average of the artifact map for each frame, as shown in Eq. (14).

$$score = \overline{\|C\|_1} \tag{14}$$

Then, whether the test image is real or fake is judged by comparing the artifact map score and the decision threshold. That is, when the artifact map score is greater than the threshold, the test image is classified as fake, as shown in Eq. (15). The decision threshold during the testing phase is set to 0.1, which is an experience value.

$$\begin{cases} if\ \ score \geq threshold, then\ \ y = 0\ (fake) \\ if\ \ score < threshold, then\ \ y = 1\ (real) \end{cases} \tag{15}$$

Table 2
Comparison with baselines on dataset II.

Methods	Deepfake	Face2Face	Faceswap
LGSC [32]	89.45%	90.30%	91.72%
FaceNet [37]	94.85%	95.64%	93.69%
XceptionNet [4]	95.45%	96.98%	94.67%
DAGFD	**97.03%**	**97.82%**	**95.97%**

5. Experiment results and analysis

In this section, the extensive experimental results are presented to demonstrate the effectiveness and superiority of the proposed DAGFD.

5.1. Comparative experiment

To validate the effectiveness, the proposed method is compared with LGSC [32] and XceptionNet [4]. The reason why the proposed method compares with LGSC and XceptionNet is that DAGFD is proposed based on the aforementioned two classical methods, which aims to improve the generalization ability. In addition, FaceNet [37] which uses metric loss aiming to locate the forgery regions is also compared.

Table 2 presents the experimental results on Deepfake, Face2Face and Faceswap on dataset II. The results show that for all manipulation types, the accuracy has been improved. For Deepfake and Faceswap, the accuracy of DAGFD has been improved significantly, which has increased by more than 1% compared with XceptionNet. On the other hand, the accuracy improvement of Face2Face type is less, about 0.9%. The reason why the increase is small is that the Face2Face type is to manipulate expressions and hence the forgery area is small, which weakens the advantages of the proposed method.

5.2. Performance comparison with state-of-art methods

To demonstrate the superiority of DAGFD over existing state-of-the-art methods, comparative experiments are conducted on all manipulation types of dataset I. Table 3 compares the accuracy of DAGFD with several state-of-the-art methods.

It can be seen from Table 3 that compared with state-of-the-art methods, the detection performance of DAGFD has improvement. Compared with the method proposed by Rossler et al. [4], the accuracy has an increase of about 2.3%. Compared with the methods proposed in [6,7,30], the proposed method is also better. Furthermore, the error bar on ACC of DAGFD is provided, and the result demonstrates DAGFD has comparable performance to BiG-Arts [16]. In summary, DAGFD outperforms state-of-the-art detection methods by leveraging the strengths of AMD and AC. AMD enhances detection by locating forgery regions and amplifying artifact features, while AC benefits feature learning with high flexibility in optimization.

Table 3
Comparison of accuracy between DAGFD and state-of-the-art methods on dataset I.

Methods	All(c23)
Xception [4]	95.73%
Li et al. [6]	96.69%
Zhao et al. [7]	97.60%
Wang et al. [30]	97.29%
Chen et al. [16]	**99.39**%
DAGFD	98.08% (±1.05)

5.3. Generalization ability evaluation

Cross-manipulation evaluation. To demonstrate the generalization ability of the proposed method, cross-manipulation evaluation is conducted. Specifically, cross-manipulation evaluation is trained on dataset generated by one manipulation method and then tested on the one generated by the other manipulation method. This is a challenging scenario as the details of forgery samples generated from different manipulation methods vary significantly.

The proposed DAGFD is compared with state-of-the-art methods, including XceptionNet [4], Luo et al. [38], Zhao et al. [7], Huang et al. [26], and DAGFD-B, where DAGFD-B represents that the proposed DAGFD uses binary classifier, rather than artifact map scores for forgery detection. XceptionNet serves as a baseline representing classical CNN classifier. The experimental results are shown in Table 4, where **bold** indicates the best performing method, and *italic* font indicates the second best method. Firstly, the experimental results of XceptionNet [4] demonstrate that the binary classification methods based on CNN have a significant performance degradation on the forgery videos generated by unseen manipulation methods. Namely, XceptionNet achieves excellent performance on seen forgery type, while the performance on unseen forgery types will degrade dramatically. The reason for the poor generalization is the pursuit of high accuracy leads to the overfitting phenomenon, and the diverse generation methods of forgery videos make it difficult for the model to form an obvious decision boundary in the feature space. Furthermore, the results show that compared with XceptionNet, DAGFD's accuracy is increased by more than 10% when testing on the dataset of different manipulation types from the training set, and the generalization ability is better.

In addition, DAGFD has better or comparable performance compared with state-of-the-art methods. However, DAGFD is inferior to Huang et al. [26] when trained on FaceSwap and tested on DeepFake. The reason is that the work in [26] is specifically designed for face swapping using face identity. When trained on Face2Face and tested on FaceSwap, the methods in [7,38] surpass DAGFD. This is because the work in [38] utilizes high-frequency features, and the work in [7] aggregates textural features and semantic features, namely comparative methods have richer feature representations. And in most cases, DAGFD outperforms DAGFD-B, indicating that the generalization of supervised anomaly detection based method is better than that of classifier based method.

Generalize from FF++ to DFD. To further demonstrate the generalization ability of the proposed method, cross-dataset experiments are conducted. In this experiment, we train models on FF++ (HQ) and evaluate them on DFD [39]. This setting is challenging since the testing sets share much less similarity with the training set. We can see from Table 5 that our method achieves apparent improvements over XceptionNet [4], BiG-Arts [16] and Luo et al. [38] about 11.6%, 4.79%, and 2.81%, respectively. Moreover, when the models are trained with NeuralTextures, DAGFD outperforms DAGFD-B by 7.64%. This further highlights the superior generalization capability of supervised anomaly detection method compared to classifier based method.

In summary, DAGFD has generalization ability improvement, and the reasons are as follows: (1) The artifact map score is proposed from the perspective of supervised anomaly detection, where the real face score tends to close to 0, while the one of the forgery face is far greater than 0. In this way, the proposed method can generalize well to unseen forgery methods; (2) With MSE loss, the artifact map of forgery sample is constrained to all-one mask regardless of whether the forgery sample is seen or not, thus improving the detection accuracy of unseen forgery types; (3) Triplet focal loss combining with metric-softmax loss, where triplet focal loss is used as the regularizer, improves the generalization ability. The validity will be experimentally demonstrated in Section 5.5.2.

Table 4
Cross-manipulation evaluation on FF++.

Training set	Method	Testing set (%)		
		DeepFake	FaceSwap	Face2Face
DeepFake	XceptionNet [4]	95.45	42.33	51.71
	Zhao et al. [7]	**99.92**	40.61	75.23
	Luo et al. [38]	*99.87*	47.21	**76.89**
	Huang et al. [26]	99.51	63.83	–
	DAGFD-B	99.65	*63.89*	74.71
	DAGFD	99.72	**65.21**	*75.36*
FaceSwap	XceptionNet [4]	49.55	95.11	48.76
	Zhao et al. [7]	64.13	*99.67*	66.39
	Luo et al. [38]	70.21	**99.85**	68.72
	Huang et al. [26]	**75.39**	98.86	–
	DAGFD-B	72.44	99.02	*71.45*
	DAGFD	*73.79*	99.15	**71.77**
Face2Face	XceptionNet [4]	52.04	50.27	95.69
	Zhao et al. [7]	86.15	*60.14*	98.66
	Luo et al. [38]	*89.23*	**61.3**	98.77
	Huang et al. [26]	–	–	–
	DAGFD-B	**89.82**	55.73	*98.82*
	DAGFD	88.91	59.83	**98.93**

Table 5
Cross-dataset Comparison from FF++ to DFD.

Training set	Test set	Detection accuracy (ACC)				
		Xception	BiG-Arts [16]	Luo [38]	DAGFD-B	DAGFD
FF++	DFD	83.1%	89.92%	91.9%	93.86%	**94.71**%
NeuralTextures	DFD	–	–	–	84.15%	**91.79**%

5.4. Ablation experiment

To evaluate the contributions of DAGFD's individual components, the following ablation experiments are carried out on dataset II: (1) experiment 1: the baseline configuration, where triplet loss, softmax loss, and ResNet18 are applied; (2) experiment 2: compared with experiment 1, the batch size changes from 32 to 64; (3) experiment 3: compared with experiment 2, triplet focal loss is used instead of triplet loss; (4) experiment 4: compared with experiment 3, XceptionNet replaces ResNet18; (5) experiment 5: compared with experiment 4, circle loss replaces softmax; (6) experiment 6: compared with experiment 5, metric-softmax loss is added; (7) experiment 7: compared with experiment 6, MSE loss is added. The results are shown in Table 6. From experiment 1 and 2, we can conclude that larger batch size helps to improve the detection accuracy by 3.4%. From experiment 2 and 3, triplet focal loss improves the performance by 1.36%, which makes hard samples to be classified correctly. From experiment 3 and 4, the performance improvement is 1.65%, which illustrates the features extracted by XceptionNet have strong representation ability. Compared with experiment 4, the accuracy of experiment 5 increases, which means circle loss can make the model self-adaptive optimization. From experiment 5 and 6, the introduction of metric-softmax loss improves the performance, which promotes the model to learn more discriminative features to distinguish real samples from fake samples. Finally, experiment 7 is the proposed method with highest accuracy, where MSE loss is used to enlarge the difference between artifact maps of the real samples and fake samples.

Table 6
Ablation experiment on dataset II.

	Batch size	Triplet loss	Triplet focal loss	MSE loss	Metric-softmax loss	Xception	ResNet18	Softmax	Circle loss	ACC on DeepFake
1	32	√					√	√		89.45%
2	64	√					√	√		92.86%
3	64		√				√	√		94.22%
4	64		√			√		√		95.87%
5	64		√			√			√	96.12%
6	64		√		√	√			√	96.45%
7	64		√	√	√	√			√	**97.03%**

Table 7
The effect of metric-softmax loss.

Methods	FF++	DFD
XceptionNet [4]	95.73%	83.1%
DAGFD w/o Metric-softmax	95.87%	91.36%
DAGFD	**98.08%**	**94.71%**

Table 8
The effect of triplet focal loss.

Methods	FF++	DFD
XceptionNet [4]	95.73%	83.1%
DAGFD w/o Triplet Focal Loss	96.75%	89.96%
DAGFD (Triplet Loss)	96.88%	92.47%
DAGFD	**98.08%**	**94.71%**

Table 9
The effect of MSE loss.

Methods	FF++	DFD
XceptionNet [4]	95.73%	83.1%
DAGFD w/o MSE	96.93%	90.82%
DAGFD	**98.08%**	**94.71%**

Table 10
The effect of circle loss.

Methods	FF++	DFD
XceptionNet [4]	95.73%	83.1%
DAGFD (Softmax)	96.38%	91.2%
DAGFD	**98.08%**	**94.71%**

5.5. The effect of diverse losses

Further ablation studies are conducted to evaluate the effect of diverse losses on in-dataset and cross-dataset ability. In the following section, we train all models on FF++ [4] and evaluate their performance on FF++ and DFD [39].

5.5.1. The effect of metric-softmax loss

Table 7 shows the effect of metric-softmax loss on in-dataset and cross-dataset evaluation. The evaluated variants include the baseline XceptionNet, the proposed DAGFD without metric-softmax loss, and DAGFD. From Table 7, the accuracy of DAGFD has improvement compared to DAGFD without metric-softmax on FF++ and DFD, which is about 2.21% and 3.35%, respectively. The reason for the improvement is that metric-softmax loss promotes to obtain discriminative feature representations.

5.5.2. The effect of triplet focal loss

To evaluate the impact of triplet loss and triplet focal loss on in-dataset and cross-dataset ability, the ablation study is conducted. The results are reported in Table 8. The evaluated variants include the baseline XceptionNet, DAGFD without triplet focal loss, DAGFD with triplet loss, and DAGFD. Regarding the ablation study, we observe that triplet focal loss performs better than triplet loss by 1.2% on FF++, demonstrating that triplet focal loss helps to mine hard negative samples to improve performance. Furthermore, the performance of DAGFD without triplet focal loss is 4.75% worse than that of DAGFD on DFD, which means triplet focal loss combining with metric-softmax loss, as the regularizer, improves the generalization ability.

5.5.3. The effect of MSE loss

Table 9 presents the impact of MSE loss on in-dataset and cross-dataset evaluation. The evaluated variants include XceptionNet, the proposed DAGFD without MSE loss, and DAGFD. From Table 9, DAGFD outperforms DAGFD without MSE by 3.89% on DFD. The experimental results demonstrate that pixel-wise supervision helps to constrain the artifact map of forgery sample to all-one mask regardless of whether the forgery sample is seen or not, achieving better generalization performance.

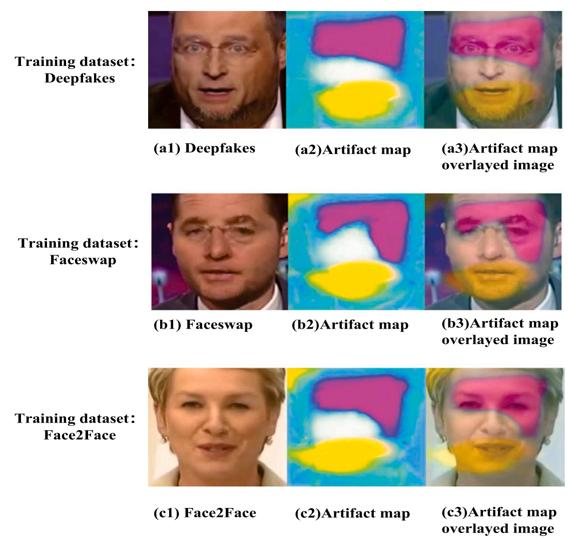

Fig. 4. Visualization of the artifact map.

5.5.4. The effect of circle loss

Table 10 demonstrates the influence of circle loss on in-dataset and cross-dataset evaluation. The evaluated variants include XceptionNet, DAGFD with softmax loss, and the proposed DAGFD with circle loss. Compared with DAGFD with softmax loss, the proposed DAGFD has an increase of 1.7% and 3.51% on FF++ and DFD, respectively. This is due to the fact that circle loss has circular decision boundary and optimization flexibility, which is superior to Softmax-based classification loss.

5.6. The effect of auxiliary classifier

Table 11 shows the effect of auxiliary classifier on in-dataset and cross-dataset evaluation, where the model is trained on FF++. The evaluated variants include the baseline XceptionNet, the proposed DAGFD trained without auxiliary classifier, and DAGFD. From Table 11, the accuracy of DAGFD has improvement compared to DAGFD without auxiliary classifier on FF++ and DFD, which is about 1.29% and 5.09%,

Fig. 5. Feature distribution of XceptionNet (left) and DAGFD (right).

Fig. 6. Failure cases.

Table 11
The effect of auxiliary classifier.

Methods	FF++	DFD
XceptionNet	95.73%	83.1%
DAGFD w/o AC	96.79%	89.62%
DAGFD	**98.08%**	**94.71%**

respectively. The reason for the improvement is that auxiliary classifier assists in obtaining discriminative artifact features, which provides additional constraints for parameter updating of AMD.

5.7. The effect of perturbations on DAGFD

To evaluate the robustness of proposed method, we conducted experiments analyzing its performance under different perturbations, including Gaussian noise, Gaussian blur, and compression. Specifically, the model is trained on FF++ [4] and tested on FF++ and DFD [39], with perturbations implemented using the Albumentations library. Perturbations are characterized by three key parameters: noise variance, kernel size, and image quality. For the low image quality condition, the parameters are set to 0.3, 7×7, and 30, respectively. As shown in Table 12, the experimental results indicate that the proposed method is robust to Gaussian noise. Furthermore, while performance shows a slight decrease at low levels of Gaussian blur and compression, significant degradation is observed at high levels.

5.8. Time performance comparison

In order to demonstrate that DAGFD is more suitable for practical applications, we compare the number of network parameters and processing time of different methods on dataset II. The results are shown in Table 13. It can be concluded that the processing time for a single image of our method is about 3 times faster than XceptionNet [4] and about 4 times faster than LGSC [32]. In summary, the proposed method only uses the artifact map detector to detect face images during inference, and AC module is removed. Therefore, DAGFD achieves high accuracy with few parameter number and fast processing time and is more suitable for practical applications.

Table 12
Perturbation evaluation of DAGFD on FF++ and DFD.

Perturbation type	Perturbation level	Testing set (%)	
		FF++	DFD
Gaussian noise	w/o noise	98.08	94.71
	0.1	98.04	92.92
	0.2	97.95	92.4
	0.3	97.96	91.64
Gaussian blur	w/o blur	98.08	94.71
	3 × 3	97.16	90.56
	5 × 5	94.52	85.6
	7 × 7	85.84	72.4
Compression	w/o compression	98.08	94.71
	90	97.32	92.8
	70	96.96	91.48
	50	94.36	85.2
	30	76	66.72

Table 13
Comparison of the number of network parameters and processing time between different methods.

Methods	Accuracy(ACC)	Parameter	Single image detection time
XceptionNet [4]	95.45%	20,811,050	0.0185 s
LGSC [32]	89.45%	26,901,449	0.0231 s
DAGFD	**97.03%**	**6,090,399**	**0.0058 s**

5.9. Visualization

5.9.1. Visualization of the artifact map

For a more intuitive understanding, the artifact map is visualized in Fig. 4. The first, second, and third row represents Deepfakes, Faceswap, and Face2Face, respectively, and the first, second, and third column is the input image, artifact map, and the fusion results of the artifact map and the input image.

Fig. 4 shows that the artifact map can locate the forgery regions when a fake face image is input, regardless of the forgery type. Obviously, the use of MSE loss can make the location information of manipulation regions more salient, thus the artifact map score difference between real and fake images is larger, which is more helpful for generalization to unseen generation methods.

5.9.2. Feature distribution visualization

We visualize the feature distributions of XceptionNet and DAGFD in the FF++ dataset using t-SNE, as illustrated in Fig. 5. The visualization results reveal notable differences in intra-class and inter-class feature distance between XceptionNet and DAGFD. Compared with

XceptionNet, DAGFD promotes intra-class aggregation, and expands the inter-class variance, ultimately improving the model's generalization ability.

5.9.3. Failure cases

Fig. 6 illustrates the failure cases on FF++ (c40) [4], which has high compression rate. The experimental results demonstrate performance improvement of DAGFD on compressed forgery video is limited. From Fig. 6, we can observe that the artifact is minor, and hence artifact map score is less than the given threshold, causing forgery video misclassification. This is due to the fact that high compression rate leads to low image quality, and hence fine-grained artifacts in fake videos, such as edges of eyes, may be removed after applying video compression, which can cheat the detector to judge the fake video as real.

6. Conclusion

From the perspective of supervised anomaly detection, this paper proposes the algorithm for generalized forgery detection, including the artifact map detector and the auxiliary classifier. Specifically, MSE loss is utilized to supervise the differences between the artifact map and the masks, making the model constrain the artifact map of the real sample and enhance the one of the fake sample. Then, the artifact map score is calculated in the testing phase to determine whether the video has been forged. Furthermore, auxiliary classifier is used to assist the artifact map detector module. Finally, the experimental results show that the DAGFD has performance and generalization improvement.

The main advantages of the proposed methods may be as follows: (1) The method is proposed from the perspective of supervised anomaly detection, and the classifier is utilized as an auxiliary module. (2) Instead of using traditional classifiers, the artifact map score is calculated to judge whether the video has been forged. (3) The generalization performance of the proposed method has been improved.

One weakness is that the performance improvement of the proposed method on the compressed forgery video is limited. To address this limitation, one potential future direction is to jointly consider video encoding and decoding and forgery detection.

CRediT authorship contribution statement

Fan Zhang: Writing – review & editing, Writing – original draft, Methodology. **Ting Yang:** Software, Data curation. **Lin Cao:** Resources, Funding acquisition. **Kangning Du:** Formal analysis, Conceptualization. **Yanan Guo:** Visualization, Validation. **Peiran Song:** Formal analysis, Conceptualization. **Chen Shao:** Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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